Project Proposal: Latent-Space Data Assimilation for Sparse-Conditioned Video Prediction

Anonymous Author(s)
Affiliation
Address
email

Abstract

This proposal studies whether data assimilation (DA) can improve sequential video prediction when only sparse observations are available after the observed prefix. The central idea is that long-horizon video prediction is an unstable extrapolation problem: model errors accumulate unless the prediction is repeatedly re-anchored to reality. We propose to formulate video rollout as a latent-space forecast–analysis process, train a lightweight open-loop predictor as the DA host model, and apply latent ensemble Kalman updates whenever sparse observations become available. The initial study will focus on Moving MNIST with simulated noisy observations derived from held-out future frames. The project will evaluate both prediction quality and DA-specific uncertainty metrics, including RMSE, PCC, entropy, relative entropy, and mutual information. The main expected outcome is a clear demonstration of how assimilation changes the accuracy and uncertainty of sequential video prediction, together with a principled discussion of when DA helps and when it becomes computationally expensive.

1 Motivation

Sequential prediction problems become increasingly unstable as the rollout horizon grows. In open-loop video prediction, each forecasted frame becomes the input for the next step, so local prediction errors compound over time. This makes long-horizon prediction qualitatively different from interpolation or short-range forecasting: the farther the prediction extends beyond the observed prefix, the less constrained it is by reality.

Data assimilation offers a natural response to this instability. Rather than extrapolating indefinitely, DA periodically corrects the forecast whenever new observations arrive. From this perspective, stable prediction of a known world should not rely on unconstrained extrapolation alone; it should instead combine model-based forecasting with repeated re-anchoring to sparse evidence. This proposal asks whether that philosophy can be made concrete for video prediction in a small, controlled machine-learning project.

2 Problem Statement

Let $x_t \in \mathbb{R}^{H \times W}$ be the video frame at time t and z_t be its latent representation. The project will study sparse-conditioned video prediction in which:

1. the first T_c frames are observed as context,
2. the next T_p frames must be predicted sequentially,
3. sparse observations of future frames may arrive intermittently during rollout.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

The latent forecast model is written as

$$z_{t+1}^{f} = M_{\theta}(z_t^{a}) + \eta_{t+1}, \tag{1}$$

where M_θ is a learned latent dynamics model and η_{t+1} is forecast uncertainty. Observations are modeled as

$$y_{t+1} = H(x_{t+1}) + \epsilon_{t+1}, \tag{2}$$

where H is the observation operator and ϵ_{t+1} is observation noise. The analysis step updates the latent forecast through

$$z_{t+1}^{a} = z_{t+1}^{f} + K_{t+1}\left(y_{t+1} - H(D(z_{t+1}^{f}))\right), \tag{3}$$

with Kalman gain estimated from a latent ensemble.

The core research question is:

> Can latent-space data assimilation improve the quality and uncertainty behavior of sparse-conditioned video prediction compared with open-loop rollout of the same predictor?

3 Why This Is an ML Project

This project sits at the intersection of machine learning and control-inspired state estimation:

1. the forecast model is a learned neural video predictor,
2. the state representation is a learned latent space,
3. the correction mechanism is imported from DA and adapted to a modern ML setting.

The machine-learning aspect is therefore not only the use of a neural network. It is also the study of how learned latent representations interact with uncertainty-aware sequential inference. The proposed experiments ask whether a learned predictor can be made more stable by combining it with structured probabilistic updates rather than relying on purely feed-forward or recurrent rollout alone.

4 Proposed Method

4.1 Reference Baseline and DA Host

The project will use two model roles:

1. a literature-aware reference baseline such as SimVP [1],
2. a lightweight open-loop latent predictor that serves as the DA host model.

The DA host model will consist of:

- a convolutional encoder,
- a ConvLSTM latent dynamics module,
- a residual latent prediction head,
- a convolutional decoder.

Given context frames $x_{1:T_c}$, the model encodes them into latent feature maps and builds a recurrent state. It then predicts future latent states autoregressively:

$$h_{t+1}, c_{t+1} = \mathrm{ConvLSTM}(z_t, h_t, c_t), \tag{4}$$
$$\hat{z}_{t+1} = z_t + G(h_{t+1}), \tag{5}$$
$$\hat{x}_{t+1} = D(\hat{z}_{t+1}). \tag{6}$$

4.2 Observation Design

The most important modeling decision is the observation source. In real-world long-horizon video generation, there may be no obvious future observation stream. To study the DA mechanism in a controlled and defensible way, the initial project will use *simulated observations* generated from held-out ground truth at test time. This is standard in DA benchmarking and is not considered cheating as long as:

1. the future truth is not used during training,
2. only partial/noisy observations are revealed,
3. the model must still infer the unobserved future structure.

The initial observation regimes will include:

1. full-frame observations every k future steps,
2. masked-pixel observations,
3. low-resolution frame observations,
4. additive Gaussian observation noise.

4.3 Latent ETKF

At each assimilation step, an ensemble of latent forecast states will be propagated through the host model. The predicted frames will be projected into observation space and compared with the available observation. An ETKF-style update will then correct the latent ensemble before the rollout continues.

This is attractive for three reasons:

1. latent space is much lower-dimensional than pixel space,
2. ensemble updates provide a natural uncertainty estimate,
3. the forecast–analysis structure matches the DA interpretation exactly.

5 Experimental Plan

5.1 Dataset

The primary benchmark will be Moving MNIST, using:

- grayscale frames of size 64×64,
- two moving digits per sequence,
- 20 frames per sequence,
- prediction task: first 10 frames $\rightarrow$ next 10 frames.

This dataset is intentionally simple. The goal is not to claim realism, but to make the DA mechanism transparent and fast enough to implement, debug, and analyze within the project timeline.

5.2 Baselines

The main comparisons will be:

1. SimVP reference baseline,
2. open-loop DA host without any future assimilation,
3. DA host with always-assimilate latent ETKF,
4. optional selective-assimilation extension if time permits.

5.3 Evaluation Metrics

The project will report the required DA metrics:

1. RMSE,
2. PCC,
3. entropy,
4. relative entropy,
5. mutual information.

Their purpose is different:

- RMSE and PCC measure prediction quality,
- entropy measures forecast uncertainty,
- relative entropy measures how strongly the observation changes the forecast,
- mutual information measures how much uncertainty the observation removes.

Runtime and number of assimilation events will also be recorded, because a longer-term motivation of the project is to understand the quality–compute tradeoff.

6 Expected Outcomes

The main expected outcome is that latent ETKF will improve prediction quality relative to the same open-loop host model under sparse observations. More specifically, the project expects to show:

1. lower RMSE and higher PCC after assimilation,
2. lower posterior uncertainty compared with the open-loop forecast,
3. interpretable entropy / relative entropy / mutual information values that reflect a genuine forecast–analysis correction process.

The project does *not* assume that always-assimilate ETKF will immediately be faster than open-loop prediction. In fact, a likely outcome is that ETKF improves quality but costs more runtime. This would still be a useful finding, because it would motivate a later selective-assimilation extension in which corrections are applied only when innovation, spread, or entropy is sufficiently large.

7 Planned Deliverables

By the end of the project, the intended deliverables are:

1. a trained SimVP reference baseline on Moving MNIST,
2. a trained open-loop latent predictor that supports stepwise rollout and assimilation,
3. a latent ETKF implementation with configurable sparse observation operators,
4. an experimental comparison of open-loop and assimilated prediction using RMSE, PCC, entropy, relative entropy, and mutual information,
5. a final report that explains the DA formulation, documents the experimental setup, and discusses both strengths and limitations of the approach.

These deliverables are intentionally chosen so that the project remains feasible within a course timeline while still producing a result that is methodologically complete: there is a learned model, a DA mechanism, a controlled benchmark, and a quantitative uncertainty analysis.

Table 1: Planned execution stages for the ML project.

Stage	Goal
1	Train and validate the SimVP reference baseline
2	Train the open-loop DA host predictor
3	Implement latent ETKF with simulated sparse observations
4	Compare open-loop and ETKF under the same host checkpoint
5	Measure entropy, relative entropy, and mutual information
6	Extend to richer observation regimes or selective assimilation

8 Risks and Limitations

The project has several clear risks:

1. the open-loop host may be too weak or too blurry,
2. latent ETKF may improve quality only slightly,
3. the observation setup may be criticized if it is not framed carefully,
4. the method may improve quality without improving compute.

These risks will be addressed by:

1. keeping the benchmark controlled and transparent,
2. explicitly treating the observation stream as a synthetic DA benchmark,
3. comparing DA against the same open-loop host rather than changing multiple things at once,
4. discussing quality improvement and runtime cost separately.

9 Milestones

10 Conclusion

This proposal frames latent-space data assimilation as a machine-learning project about stabilizing sequential prediction with structured probabilistic updates. The project is deliberately scoped to a clean benchmark so that the key ideas remain analyzable: a learned latent predictor, sparse observations, ETKF correction, and uncertainty-aware metrics. If successful, the project will provide a concrete demonstration that DA can improve sparse-conditioned video prediction and offer a principled way to think about long-horizon forecasting as repeated local re-anchoring rather than unconstrained extrapolation.

References

[1] Zhangyang Gao, Cheng Tan, Lirong Wu, and Stan Z. Li. Simvp: Simpler yet better video prediction. *arXiv preprint arXiv:2206.05099*, 2022. doi: 10.48550/arXiv.2206.05099.